

Coding Animal Health+ AMR laboratory function competency

Using the Fleming Fund 'Site Planning and Monitoring Forms'

This guide demonstrates how Fleming Fund site form questions are used to code the level of competency of AMR sentinel site functions for each laboratory (Supplementary Figure 1).

By following this guide, one can better understand the question answers and how they affect functional competency.

Component 1: Sampling procedure and investigation

Subcomponent 1a: Clinical samples and samples collected as part of passive surveillance activities - Question A2.1.4

Core

- To achieve 'Core', relevant demographic and epidemiological data must be collected for each sample. A unique alphanumeric identifier must also be used for the sample and matching epidemiological/demographic data. Paper records may be used, and data protection (confidentiality) and ethics guidelines must be in place and followed.
- As such, sites must answer 'Yes' to question A2.1.4:
 - 'Does the site have any tools, guidelines or SOPs to capture demographic or epidemiological data when receiving a new sample?'
- If the site is answering after 2024*, they must also answer 'Paper log books', 'On local computers', 'Using WHONET', or 'LIMS system other than WHONET' to question B6.1.1:
 - 'How are laboratory results stored?'
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- Sites must finally also answer 'Yes' to B6.4.2:
 - 'Do the records include unique alphanumeric ID to link bacteriology and demographic/epidemiology data?'

Extended

- To achieve 'Extended', data must be collected using electronic technology. Further epidemiological/demographic data may be collected as samples are taken to carry out more sophisticated analysis. Information is must also be fed back to farmers,

prescribers and sellers to create workshops to raise awareness about the threats posed by AMR and provide them with alternatives to antimicrobial consumption.

- Therefore, If the site is answering after 2024*, they must answer ‘On local computers’, ‘Using WHONET’, or ‘LIMS system other than WHONET’ to question B6.1.1.
- In addition, sites must answer ‘Yes’ to question B6.3.3
 - ‘Are any guidelines in place for sharing results with the sample source (i.e. farmers, vets)?’

Question	Core	Extended
A2.1.4	Yes	Yes
B6.1.1 (2024-onwards)	Paper log books/On local computers/Using WHONET/LIMS system other than WHONET	On local computers/Using WHONET/LIMS system other than WHONET
B6.3.3	No	Yes
B6.4.2	Yes	Yes

Subcomponent 1b: Training and quality assurance - Questions A2.1.2, A2.1.5, A2.2.1-A2.2.2, B4.7 and B4.9

Core

- To achieve 'Core', there must be routine training for surveillance sampling SOPs. Furthermore, there must be quality control and Internal Quality Assessment of sample and data collection, including linkage of diagnostic results and demographic data.
- As such, sites must answer 'Yes' to questions A2.1.2 and A2.1.5:
 - A2.1.2: 'Have SOPs been developed for AMR-specific active surveillance sampling?'
 - A2.1.5: 'Has an SOP been developed for internal quality control of sampling processes?'
- Furthermore, sites must answer 'All staff' to questions A2.2.1 and A2.2.2:
 - A2.2.1: 'Have all relevant laboratory staff received training for AMR surveillance-specific SOPs?'
 - A2.2.2: 'Have relevant laboratory staff received up to date training on sampling procedures (biosafety, sample handling and integrity, sample acceptance/rejection criteria etc)?'
- Together, the answers to these questions show that relevant SOPs exist, and that relevant staff are trained on them.

Extended

- To achieve 'Extended', sites must also be involved in an External Quality Assessment programme.
- As such, sites must answer 'Yes' to question B4.7:
 - 'Is the laboratory participating in an accredited External Quality Assessment scheme for bacterial identification and antimicrobial susceptibility testing?'

Advanced

- To achieve 'Advanced', sites must also act as a regional training centre.
- As such, sites must answer 'Yes' to question B4.9:
 - 'Does the laboratory provide EQA to other sites?'

Question	Core	Extended	Advanced
A2.1.2	Yes	Yes	Yes
A2.1.5	Yes	Yes	Yes
A2.2.1	All staff	All staff	All staff
A2.2.2	All staff	All staff	All staff
B4.7	No	Yes	Yes
B4.9	No	No	Yes

Component 2: Isolate identification and susceptibility testing

Subcomponent 2a: Sample transport - Questions A3.1-A3.3

Core/Extended

- To achieve 'Core', samples must be transported according to local SOPs. To achieve 'Extended', samples must be transported according to international biosafety standards.
 - The questions in the AH site form ask about international biosafety standards, therefore 'Core' is skipped straight to 'Extended'
- As such, to achieve 'Extended', sites must answer 'Yes' to question A3.1:
 - 'Have SOPs been developed for sample transport to the laboratory in line with international biosafety standards?'
- In addition, sites must also answer 'All staff' to questions A3.2 and A3.3:
 - A3.2: 'Have all relevant laboratory staff received up to date training for the sample transport SOPs?'
 - A3.3: 'Are staff involved in sample collection in the field following the SOPs to ensure that samples are transported safely?'
- This ensures that samples are transported properly at all stages of handling.

Question	Extended
A3.1	Yes
A3.2	All staff
A3.3	All staff

Subcomponent 2b: Culture and identification - Questions B2.1.1-B2.1.6 & B2.1.8

Core

- To achieve 'Core', sites must have the capacity to identify the relevant priority pathogens according to SOPs.
- Sites must therefore answer 'Yes, according to SOP' to questions B2.1.1 and B2.1.2. If the site is answering after 2024**, the answer must also be provided to question B2.1.3.
 - 'Please indicate if the laboratory can perform culture, ID and AST for any of the following organisms'

Required –

- B2.1.1: *Escherichia coli*
- B2.1.2: *Salmonella* spp.

Required 2024-onwards –

- B2.1.3: *Staphylococcus aureus*

Extended

- To achieve 'Extended', sites must have the capacity to identify an extended range of pathogens according to SOPs.
- Therefore, as well as the requirements for 'Core', sites must therefore answer 'Yes, according to SOPs' to question B2.1.5:
 - *Enterococcus* spp.

Advanced

- To achieve 'Advanced', sites must perform automated identification (e.g. Vitek2, BD Phoenix, Microscan, MALDI TOF).
- In addition to the requirements for 'Core' and 'Extended', sites must answer 'Yes, according to SOP' to questions B2.1.4. If the site is answering after 2024**, the answer must also be provided to question B2.1.6.

Required –

- B2.1.4: *Campylobacter* spp.

Required 2024-onwards –

- B2.1.6: *Listeria monocytogenes*
- In addition, sites must answer 'Yes, according to SOP' to question B2.1.8:
 - 'Is the laboratory using an automated system to perform bacterial identification?'

Question	Core	Extended	Advanced
B2.1.1 & B2.1.2 (& B2.1.3 2024-onwards)	Yes, according to SOP	Yes, according to SOP	Yes, according to SOP
B2.1.5	Yes, but without SOP/No	Yes, according to SOP	Yes, according to SOP
B2.1.4 (& B2.1.6 2024-onwards)	Yes, but without SOP/No	Yes, but without SOP/No	Yes, according to SOP
B2.1.8	Yes, but without SOP/No	Yes, but without SOP/No	Yes, according to SOP

Subcomponent 2c: Susceptibility testing - Questions B2.2.2 – B2.2.4

Core

- To achieve 'Core', sites must use of disc diffusion for AST of priority bacteria according to SOPs.
- Sites must therefore answer 'Disc diffusion' to question B2.2.2:
 - 'How does the laboratory perform antimicrobial susceptibility testing?'
 - Question B2.2.1 asks whether the site performs AST, but if the site answers 'Yes', they must also answer B2.2.2.
- Sites must also answer 'The latest EUCAST standard' or 'The latest CLSI standard', 'The EUCAST standard of year 20XX' or 'The CLSI standard of year 20XX' to question B2.2.4:
 - 'What standard is being used for AST?'

Extended

- To achieve 'Extended', sites must use disc diffusion for AST of an extended range of species, according to SOPs.
- Therefore, in addition to the requirements for 'Core', sites must also answer be 'Core' for subcomponent 2b, as this indicates that a range of species are being tested.

Advanced

- To achieve 'Advanced', sites must use broth dilution methods (including automated) to obtain MICs.
- In addition to achieving 'Core' for subcomponent 2c, sites must therefore also answer 'MIC testing' or 'Automated method' to question B2.2.2.
- Alternatively, sites must provide any answer other than 'Not applicable' to question B2.2.3:
 - 'How does the laboratory perform Minimum Inhibitory Concentration (MIC) testing?'
 - As both questions indicate that MIC testing is performed, either can be used to achieve 'Advanced'.

Question	Core	Extended	Advanced
B2.2.2	Disc diffusion	Disc diffusion	MIC testing/Automated method
B2.2.3	Not applicable	Not applicable	Or By Etest/By broth dilution/Using both Etest and broth dilution/Other
B2.2.4	The latest EUCAST/CLSI standard/The EUCAST/CLSI standard of year 20XX	The latest EUCAST/CLSI standard/ The EUCAST/CLSI standard of year 20XX	The latest EUCAST/CLSI standard/ The EUCAST/CLSI standard of year 20XX

Subcomponent 2b	Precore/Core	Extended	Extended
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Subcomponent 2d: Training and QA - Questions B5.1.1, B4.7 & B5.1.4

Core

- To achieve 'Core', sites must have routine training for SOPs, quality control and internal QA.
- Sites must therefore answer 'All staff' to quest B5.1.1:
 - 'Have relevant laboratory staff received training on the sample processing, organism identification and AST SOPs for the site?'

Extended

- To achieve 'Extended', sites must be enrolled in an External Quality Assessment scheme.
- As such, sites must answer 'Yes' to the previous EQA question B4.7.

Advanced

- To achieve 'Advanced', sites must also act as a regional training centre.
- As such, as well as the above, sites must also answer 'Yes' to B5.1.4:
 - 'Does the laboratory provide training for bacterial identification/AST to other sites?'

Question	Core	Extended	Advanced
B5.1.1	All staff	All staff	All staff
B4.7	No	Yes	Yes
B5.1.4	No/Not applicable	No/Not applicable	Yes

Component 3: Isolate storage (local) and referral to AMR laboratory

Subcomponent 3a: Storage of isolates - Questions B3.1.2 and B3.1.4-B3.1.7

Core

- To achieve 'Core', sites must have freezer storage (-20°C) of all isolates with linkage to paper or electronic database, or short-term storage on swabs / agar slopes.
- Therefore, sites must answer 'In -20°C freezers' to question B3.1.2:
 - 'How are bacterial isolates stored at the laboratory?'
 - In addition, 'On agar slopes / agar stabs' is also considered acceptable.
- In addition, sites must answer 'Linked via paper records / logbooks' to question B3.1.6:
 - 'If samples are stored, what method is used to link the isolate and the specimen information?'
- Finally, in order to ensure the database is kept up to date, sites must also answer 'Paper-based record / log book' to question B3.1.7:
 - 'Does the site maintain an up to date inventory of all the isolates stored?'

Extended

- To achieve 'Extended', sites must have a reliable (generator back-up) freezer storage (-80°C) of all isolates with linkage to electronic database.
- Therefore, sites must answer 'In -80°C freezers' or 'Freeze dried' to question B3.1.2.
- Furthermore, sites must answer 'Linked via electronic database' to question B3.1.6.
- Sites must also answer either 'Excel spreadsheet', 'LIMS' or 'Formal biorepository software' to question B3.1.7.
- Sites must answer 'Yes' to questions B3.1.4 and B3.1.5:
 - B3.1.4: 'If freezers are used, is the power supply reliable with a backup generator and UPS?'
 - B3.1.5: 'If freezers are used, are they under a service contract with regular external monitoring of temperature?'

Question	Core	Extended
B3.1.2	In -20°C freezers/ On agar slopes / agar stabs	In -80°C freezers/Freeze dried
B3.1.4	No/Not applicable	Yes
B3.1.5	No/Not applicable	Yes
B3.1.6	Linked via paper records / logbooks	Linked via electronic database
B3.1.7	Paper-based record / log book	Excel spreadsheet/LIMS/Formal biorepository software

Subcomponent 3b: Transport to AMR reference laboratory - Questions B3.2.1-B3.2.2

Core

- To achieve 'Core', sites must transfer relevant isolates to an AMR reference laboratory annually according to SOPs.
- Therefore, sites must answer 'Yes' to question B3.2.1:
 - 'Are isolates regularly transferred to an AMR reference laboratory?'
- Sites must also answer 'Once a year' or 'Every 3-6 months' to question B3.2.2:
 - 'How frequently are isolates transferred?'

Extended

- To achieve 'Extended', sites must transfer relevant isolates to AMR reference laboratory quarterly according to international standards for biosafety.
- Therefore, as well as 'Yes' to question B3.2.1, sites must also answer 'Every 1-3 months' to question B3.2.2.

Question	Core	Extended
B3.2.1	Yes	Yes
B3.2.2	Once a year/Every 3-6 months	Every 1-3 months

Subcomponent 3c: Training and QA - Questions B5.1.5-B5.1.6, B3.1.3 and B4.7

Core

- To achieve 'Core', sites must provide routine training for isolate storage, SOPs, quality control and internal quality assessment.
- Therefore, site must answer 'Yes' to question B5.1.5:
 - 'Are SOPs in place for sample referral and transport to an AMR reference laboratory?'
- In addition, sites must answer 'All staff' to question B5.1.6:
 - 'Have relevant laboratory staff received up to date training on these SOPs?'
- Alternatively, the site may be a national reference laboratory. As such, transportation SOPs may be irrelevant. In this instance, sites must answer 'Yes' to question B3.1.3:
 - 'Does the laboratory have SOPs for sample storage?'

Extended

- To achieve 'Extended', sites must take part in an External quality assessment scheme.
- As such, in addition to the requirements for 'Core', sites must also answer 'Yes' to the previous EQA question B4.7.

Question	Core	Extended
B5.1.5	Yes	Yes
B5.1.6	All staff	All staff
B3.1.3 (Instead of B5.1.5- B5.1.6 if site is a reference site)	Yes	Yes
B4.7	No	Yes

Component 4: Data management and review

Subcomponent 4a: Data use - Questions B6.3.1-B6.3.3 & B6.3.5

Core

- To achieve 'Core', sites must submit anonymised individual data to a national coordinating centre, and data must be reported to relevant stakeholders.
- Therefore, sites must answer 'Yes, anonymised' to question B6.3.1:
 - 'Are AMR data submitted to a national central database (or equivalent)?'
- Sites must also answer 'Annually', 'Every 3-6 months', 'Monthly' or 'Weekly' to question B6.3.2:
 - 'How frequently are data submitted?'
- Furthermore, to demonstrate that sites share data with stakeholders, sites must answer 'Yes' to the previous sample source sharing question B6.3.3.
- In addition, sites must also answer 'Yes' to question B6.3.5:
 - 'Are AMR data from this site shared with local or international stakeholders?'

Advanced

- To achieve 'Advanced', sites must have automated real time submission of data to national coordinating centre for animal health sector.
- Therefore, in addition to 'Yes' to questions B6.3.3 and B6.3.5, sites must answer 'Daily automated submission' to question B6.3.2.

Question	Core	Advanced
B6.3.1	Yes, anonymised	Yes, anonymised
B6.3.2	Annually/Every 3-6 months/Monthly/Weekly'	Daily automated submission
B6.3.3	Yes	Yes
B6.3.5	Yes	Yes

Subcomponent 4b: Data management - Questions B6.4.1-B6.4.2

Core

- To achieve 'Core', sites must store diagnostic results and sample demographic data in paper records or excel spreadsheet. Diagnostic results and sample demographic data must also be accurately linked using the unique alphanumeric identifier.
- Therefore, sites must answer 'Paper linkage' to question B6.4.1:
 - 'How are laboratory and demographic/epidemiology databases linked?'
- In addition, sites must answer 'Yes' to the previous question regarding alphanumeric B6.4.2 inclusion.

Extended

- To achieve 'Extended', sites must store diagnostic results and sample data in a customized database or LIMS, which must be regularly backed up. External quality assessment should be performed, and epidemiological/demographic data and AMR

results from the laboratories can be collated into a national database using the same unique alphanumeric identifier.

- Therefore, as well as 'Yes' to question B.6.4.2, sites must answer 'Clinical and laboratory databases use same unique patient identifier and data can be retrieved by request to IT department' to question B6.4.1.
- In addition, sites must also answer 'Yes' to the previous EQA question B4.7.

Advanced

- To achieve 'Advanced', sites must have automated linkage between clinical and laboratory databases.
- Therefore, as well as 'Yes' to questions B.6.4.2 and B4.7, sites must answer 'Automated linkage of laboratory and clinical databases' to question B6.4.1.

Question	Core	Extended	Advanced
B6.4.1	Paper linkage	Clinical and laboratory databases use same unique patient identifier and data can be retrieved by request to IT department	Automated linkage of laboratory and clinical databases
B6.4.2	Yes	Yes	Yes
B4.7	No	Yes	Yes

Subcomponent 4c: Data governance - Question B6.5.1

Core

- To achieve 'Core', sites must have data sharing policy and agreements in place in collaboration with the AMR reference laboratory and relevant ministries.
- Therefore, sites must answer 'Yes' to question B6.5.1:
 - 'Does the site have a formal AMR data sharing policy or agreement in place with relevant Ministries, other site laboratories or an AMR reference laboratory?'

Question	Core
B6.5.1	Yes

Footnotes:*** Subcomponent 1a:**

In phase 1 (before 2024) of the Fleming Fund, question B6.1.1 did not exist. Therefore, it cannot be used to determine subcomponent function.

**** Subcomponent 2b:**

In 2024, questions on new important pathogens were added to the Site Planning & Monitoring Tools. Therefore, it was determined that sites would be required to identify the important pathogen *S. aureus* to achieve 'Core', whereas *Enterococcus* spp. was deemed required for 'Extended', and *L. monocytogenes* for 'Advanced' from 2024 onwards.

Supplementary Figures:

AMR Surveillance component		Core level	Extended level	Advanced level
1. Sampling procedure and investigation.	Clinical samples and samples collected as part of passive surveillance activities.	Relevant demographic and epidemiological data are collected for each sample. A unique alphanumeric identifier is used for the sample and matching epidemiological/demographic data. Paper records may be used. Data protection (confidentiality) and ethics guidelines are in place and followed.	Data collected using electronic technology. Further epidemiological/demographic data may be collected as samples are taken to carry out more sophisticated analysis. Information is fed back to farmers, prescribers and sellers to create workshops to raise awareness about the threats posed by AMR and provide them with alternatives to antimicrobial consumption.	
	Training and quality assurance.	Routine training for surveillance sampling SOPs. Quality control and Internal Quality Assessment of sample and data collection, including linkage of diagnostic results and demographic data	External Quality Assessment.	Functions as a regional training centre.
2. Isolate identification and susceptibility testing.	Sample transport.	Samples transported according to local SOPs.	Samples transported according to international biosafety standards.	
	Culture and identification.	Capacity to identify the relevant priority pathogens according to SOPs.	Capacity to identify an extended range of pathogens according to SOPs.	Automated identification (e.g. Vitek2, BD Phoenix, Microscan, MALDI TOF)
	Susceptibility testing.	Use of disc diffusion for AST of priority bacteria according to SOPs.	Use of disc diffusion for AST of an extended range of species, according to SOPs.	Broth dilution methods (including automated) to obtain MICs.
	Training and QA	Routine training for SOPs, quality control and internal QA.	Enrolled in External Quality Assessment scheme.	Functions as a regional training centre.
3. Isolate storage (local) and referral to AMR laboratory.	Storage of isolates	Freezer storage (-20°C) of all isolates with linkage to paper or electronic database, or short term storage on swabs / agar slopes.	Reliable (generator back-up) freezer storage (-80°C) of all isolates with linkage to electronic database.	
	Transport to AMR reference laboratory.	Relevant isolates are transferred to AMR reference laboratory annually according to SOPs.	Relevant isolates are transferred to AMR reference laboratory quarterly according to international standards for biosafety.	
	Training and QA.	Routine training for isolate storage, SOPs, quality control and internal quality assessment.	External quality assessment.	
4. Data management and review	Data use.	Anonymised individual data submitted to national coordinating centre via email. Reliable prevalence estimates of resistant bacteria in species included in the active surveillance programme and resistant bacteria		Automated real time submission of data to national coordinating centre for

		in clinical cases are produced yearly and reported to relevant stakeholders.		animal health sector.
	Data management	Diagnostic results and sample demographic data stored in paper records or excel spreadsheet. Diagnostic results and sample demographic data are accurately linked using the unique alphanumeric identifier.	Diagnostic results and sample data stored in a customized database or LIMS. External quality assessment. Epidemiological/demographic data and AMR results from the laboratories can be collated into a national database using the same unique alphanumeric identifier.	Automated linkage between clinical and laboratory Databases.
	Data governance.	Data sharing policy and agreements in place in collaboration with the AMR reference laboratory and relevant ministries.		

Supplementary Figure 1. Animal health sentinel site functions for AMR surveillance. Table demonstrating four essential overall AMR surveillance components which a sentinel site must be able to perform. These are further divided into 12 subcomponent functions. Sites can perform these functions at either a level of 'Core', 'Extended', or 'Advanced', and the requirements/standards to achieve each level are detailed in the table. If a site does not meet the requirements of any of these levels, the site is designated to be functioning at a 'Precore' level for that subcomponent function.